+)))))))))))),
* INCH-POUND *
.)))))))))))-

MIL-R-83536/19
AMENDMENT 2
30 September 1996
SUPERSEDING
AMENDMENT 1
15 December 1995

MILITARY SPECIFICATION SHEET

RELAYS, ELECTROMAGNETIC, ESTABLISHED RELIABILITY, 4PDT, LOW LEVEL TO 10 AMPERES, MAGNETIC LATCH, HERMETICALLY SEALED, ALL WELDED, TRANSIENT SUPPRESSED DC COIL

This amendment forms a part of MIL-R-83536/19, dated 27 March 1992, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

FIGURE 1, side view: Delete "SERIAL NUMBER SHALL BE MARKED ON TOP OF RELAYS".

FIGURE 1, side view, dimension 1.025: Add "SQ".

PAGE 2

FIGURE 1, side view: Delete "SERIAL NUMBER SHALL BE MARKED ON TOP OF RELAYS".

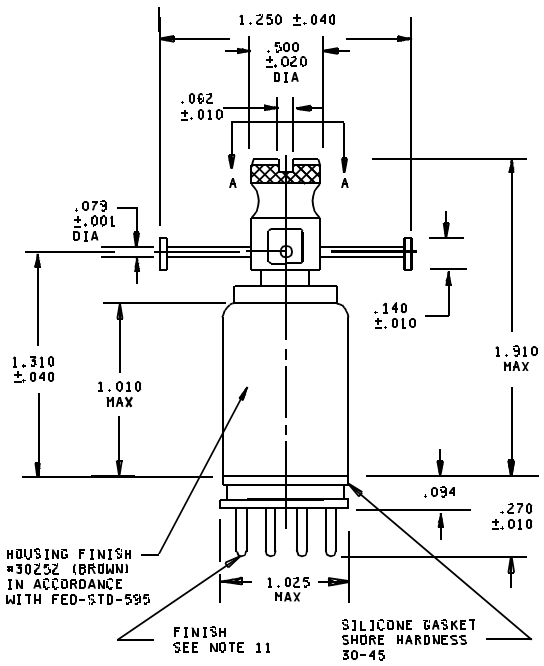
PAGE 3

* FIGURE 1, side view: Add "MARKING (SEE NOTES 7 AND 8)" and draw arrow to relay.

PAGE 6

* FIGURE 2, side view, delete and substitute the following:

"



"

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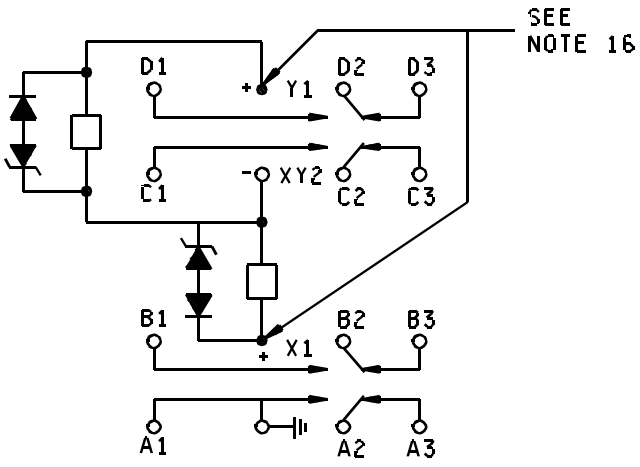
PAGE 7

FIGURE 2, add the following circuit diagram:

"



SEE NOTES 4, 5, AND 6

CIRCUIT DIAGRAM (TRACK MOUNT)
(ALTERNATE INTERNAL
DIODE CONFIGURATION)

"

PAGE 8

Delete "Contact voltage drop or resistance:" and substitute "Contact voltage drop and resistance:".

Break bounce requirement, delete in its entirety.

PAGE 9

Delete "Coil transient suppression: Applicable." and substitute "Coil transient suppression: Applicable (42 V dc maximum for track mount and 5 V dc maximum for all others)."

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TABLE III, delete and substitute:

"TABLE III. Dash numbers and characteristics. 1/

Dash number 2/		Mount	Coil data						
			At 25°C				Over temperature range	Maximum pickup voltage	
Solder hook	Socket pin		Coil voltage (V dc) 3/		Coil current (amperes) maximum	Specified latch/ reset voltage (V dc) 4/	Specified latch/ reset voltage (V dc) 4/	High temper-ature test	Contin-uous current test
			Rated	Max					
001	002	RVFM	6	7.3	0.300	3.5	4.5	5.0	5.7
003		HFM	6	7.3	0.300	3.5	4.5	5.0	5.7
004	005	RVFM	12	14.5	0.150	6.5	9.0	9.9	11.25
006		HFM	12	14.5	0.150	6.5	9.0	9.9	11.25
007	008	RVFM	28	29 5/	0.072	14.5	18.0	19.8	22.5
009		HFM	28	29 5/	0.072	14.5	18.0	19.8	22.5
	010	Track	28	29 5/	0.072	14.5	18.0	19.8	22.5
011	012	RVFM	48	50	0.038	24.0	36.0	38.0	42.0
013		HFM	48	50	0.038	24.0	36.0	38.0	42.0

1/ Each relay possesses high level and low level capabilities. However, relays previously tested or used above 10 mA resistive at 6 V dc maximum or peak ac open circuits are not recommended for subsequent use in low level applications.
2/ The suffix letter L, M, P, or R to designate the applicable failure rate level shall be L, 3.0; M, 1.0; P, 0.1; R, 0.01. Example: 001L, 002R.
3/ CAUTION: The use of any coil voltage less than the rated coil voltage will compromise the operation of the relay.
4/ Pickup voltage as shown is for test purposes only and is not to be used for design criteria.
5/ When maximum ambient temperature does not exceed +85°C, the maximum coil voltage shall be 32 V dc."

Dielectric withstanding voltage, Track mount, All other points, Initial: Delete "1000" and substitute "1500".

Dielectric withstanding voltage, Track mount, Coil to coil/Coil to case, Sea level, After life: Delete "1250" and substitute "1050".

Dielectric withstanding voltage, Track mount, Altitude, At 300,000 feet: Delete "500" and substitute "N/A" (2 places).

Maximum leakage current, delete in its entirety and substitute: "Maximum leakage current for track mount only: 100 microamperes rms."

* Delete "Shock" and all "Vibration" paragraphs and substitute:

* "Shock (specified pulse): Applicable, MIL-STD-202, method 213, test condition C, 200 g's for 6 ±1 ms, except horizontal flange mount peak g value shall be 100 g's for 6 ±1 ms. Contact chatter shall not exceed 10 microseconds maximum for closed contacts and 1 microsecond maximum closure for open contacts.

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* "Vibration (sinusoidal): Applicable, MIL-STD-202, method 204, 30 g's except frequency range shall be 10 to 3,000 Hz. Horizontal flange mount shall be 20 g's, 57 to 3,000 Hz. Contact chatter shall not exceed 10 microseconds maximum for closed contacts and 1 microsecond maximum closure for open contacts.

* "Vibration (random): Applicable to qualification and group C inspection. Test in accordance with MIL-STD-202, method 214, test condition IG (0.4 g^2/Hz, 50 to 2000 Hz); for horizontal flange mount, test condition IE (0.2 g^2/Hz) applies. Contact chatter shall not exceed 10 microseconds maximum for closed contacts and 1 microsecond maximum closure for open contacts.

* "Vibration (random): Applicable to track mount only (qualification only). Test in accordance with MIL-STD-202, method 214, test condition IG, 1.5 hours per axis. Contact chatter shall not exceed 10 microseconds maximum for closed contacts and 1 microsecond maximum closure for open contacts."

Weight: Delete "(77.2 g's)" and "(90.9 g's)" and substitute "(77.2 grams)" and "(90.9 grams)".

PAGE 13

TABLE V, second column: Delete "028" and substitute "010".

The margins of this amendment are marked with an asterisk to indicate where changes from the previous amendment were made. This was done as a convenience only, and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based upon the entire content, irrespective of the marginal notations and relationship to the last previous amendment.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Air Force - 99
DLA - ES

Preparing activity:
Air Force - 85

Agent:
DLA - ES

(Project 5945-0993-11)

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